MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Exam Type EXAMINATION Month YYYY

Semester 5

Discipline Specific Elective (DSE)- MG5DSEPHY300 –
SEMICONDUCTOR OPTOELECTRONIC DEVICES

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 35**

Record: 5 Marks

Attempt the question marked 'X'. Change of question is not allowed. (30 Marks).
Error analysis should be done wherever possible.

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

1. Study the V-I characteristics of LEDs emitting two different wavelengths and compare their turn-on voltages. [S]/[CO5]
2. Study the V-I characteristics of a LED. Find the DC and AC resistances. Also find the turn-on voltage [S]/[CO5]
3. DeterminePlanck's constant using at least 3 LEDs. [S]/[CO5]
4. Graphically determinePlanck's constant using at least 3 LEDs. [S]/[CO5]
5. Design a LED driver circuit employing a constant current source using an op-amp and transistor and study its performance. [S]/[CO5]
6. Determine the diameter of a thin wire using laser diffraction. [S]/[CO5]
7. Measure the divergence of an edge-emitting diode laser beam by measuring the dimensions of the beam projected onto a screen at different distances. [S]/[CO5]
8. Measure the diameter (beam spot size) of the laser beam. [S]/[CO5]
9. Study the optical modulation in a fibre optic communication link. [S]/[CO5]
10. Study the modulation characteristics of a Light Emitting Diode. [S]/[CO5]
11. From the given absorption/transmission data, plot the graph (using suitable software) and obtain the bandgap of a semiconductor. [S]/[CO5]
12. From the given absorption/transmission data, plot the graph (using suitable software) and obtain the absorption coefficient (of a semiconductor). [S]/[CO5]
13. Determine the wavelength of the given laser. A wire of known thickness is given. [S]/[CO5]
14. Compare the divergence of a He-Ne laser and a semiconductor laser. [S]/[CO5]
15. From the data obtained from the diffraction pattern of a thin wire using a laser, plot a graph and find the thickness of the wire. [S]/[CO5]

QP Code: _____ Reg. No.: .....

Name : .....…

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) EXAMINATION

Semester 5: MODEL QUESTION PAPER(PRACTICAL)

Computational Physics: Python

MG5DSEPHY301

Time: 2.00 Hours Maximum Marks: 30

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1. **Freely Falling Body** Write a Python program to simulate the motion of a freely falling body dropped from a height h = 100 m. Plot (i) Position Vs Time, (ii) Kinetic Energy Vs height, (iii) Potential Energy Vs height, and (iv) KE, PE, and Total Mechanical Energy Vs height on the same graph. Take mass m = 1 kg and g = 9.8 m/s². Verify the conservation of mechanical energy from the plot.

2. **Variation of Acceleration due to Gravity with Altitude** Write a Python program to calculate and plot the variation of acceleration due to gravity g with altitude (h) ranging from 0 to 1000 km above the Earth's surface. Use the values: g_0 = 9.8 m/s², radius of Earth R = 6371 km. From the plot, find the altitude at which g reduces to half its surface value.

3. **Simple Harmonic Motion:** Write a Python program to simulate simple harmonic motion of a particle of mass m = 1 kg, spring constant K = 4 N/m, and amplitude A = 1 m. Plot (i) the displacement-time trajectory, (ii) the phase space trajectory (velocityVs displacement), and (iii) variation of KE, PE, and total energy with position. Verify energy conservation from the plots.

4.**Projectile Motion** Write a Python program to simulate projectile motion for initial velocity u = 20 m/s and projection angle θ = 45°. Plot (i) the trajectory (y Vs x), (ii) the phase space plot (vy Vs vx), and (iii) KE, PE, and total mechanical energy as a function of time. Find the maximum height, range, and time of flight from the simulation and compare with analytical values.

5. **Damped Harmonic Motion** Write a Python program to numerically solve the equation of motion of a damped harmonic oscillator with mass m = 1 kg, spring constant K = 4 N/m, for three damping coefficients b = 0.5 (underdamped), b = 4 (critically damped), and b = 8 (overdamped). Plot (i) displacement Vs time for all three cases, (ii) the phase space trajectory (velocity Vs displacement), and (iii) variation of KE, PE, and total energy with position for the underdamped case. Compare the nature of oscillations in each case.

6. Determine the accuracy and processing time for different step sizes by solving algebraic equations using the Bisection and Newton-Raphson methods. Then, plot the error vs step size.

7. Solve the differential equation of a simple pendulum numerically (using the Euler and Runge-Kutta techniques), compare the result with analytical solutions, and plot the results for various initialconditions.

8. Use the Trapezoidal Rule, Simpson's 1/3-Rule, and Simpson's 3/8-Rule to fine-tune the definite integral of a given function. Then, compare the accuracy to the analytical solution. Plot the error vs. step size while repeating the experiment with various step sizes.

9. Find the maximum height of a projectile, its horizontal range, and its time of flight for varying initial velocities and projection angles.

10. Examine how the diffraction pattern varies with the slit width and wavelength of a monochromatic light source while examining diffraction patterns caused by a single slit.

11. Plot the intensity pattern for the Fresnel and Fraunhofer diffraction of monochromatic light by a single slit for different slit widths and screen distances.

12. Trace the 3-dimensional trajectory of an electron travelling in a homogeneous perpendicular electric and magnetic field

13. Examine the trajectory and phase space trajectory of a damped harmonic oscillator for various damping coefficients, (solve the differential equation numerically) and compare it with the analytical solutions.

14. Using two oscillatory functions of varying frequency and amplitude, illustrate various kinds of Lissajous figures.

15. Using the Monte Carlo method obtain the value of π(pi).

16. Using Monte Carlo technique, calculate the value of the given integral. Compare your result with the value obtained by analytical method.

17.Solve radioactive decay law to plot the number of nuclei remaining without disintegration(N) after a time t for a sample of known decay constant. From this evaluate the activity of the given sample. or similar codes suggested by the instructor.

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) MODEL QUESTION PAPER (PRACTICAL)
SEMESTER FIVE
PHYSICS OF ATMOSPHERE
(2024 ADMISSION ONWARDS)

Duration: 2 hrs **Maximum Marks: 30**

Analyse (An), Evaluate (E)

All listed practicals are examination options. The examiner shall assign one practical; the assigned practical carries 30 marks. [CO4]

1.	Analyse the variation in ground level air temperature at a location over a period of time from an automatic weather station and estimate basic statistics.	[An, E] [CO4]
2.	Analyse the variation in ground level humidity at a location over a period of time from an automatic weather station and estimate basic statistics.	[An, E] [CO4]
3.	Analyse the variation in ground level wind speed and direction at a location over a period of time from an automatic weather station and estimate basic statistics.	[An, E] [CO4]
4.	Analyse the variation in rainfall at a location over a period of time from an automatic weather station and estimate basic statistics.	[An, E] [CO4]
5.	Analyze the air quality index at a location from the Central Pollution Control Board and monitor air quality over a period of time.	[An, E] [CO4]
6.	Study and interpret meteorological charts (identification of isobaric patterns, signs and symbols) from the India Meteorological Department (IMD).	[An, E] [CO4]
7.	Analyze monsoon charts from the India Meteorological Department (IMD) and identify the features/trends.	[An, E] [CO4]
8.	Using radiosonde data over any location, analyse the vertical profile of temperature and estimate various geophysical parameters (tropopause height, boundary layer height, lapse rate) over a period of time.	[An, E] [CO4]
9.	Using radiosonde data over any location, analyse the vertical profile of humidity to identify the cloud layers for a period of time and compute their statistics.	[An, E] [CO4]
10.	Estimate convective available potential energy and convective inhibition, and analyse them for different events (clear days, cloud days, cyclone events, etc.).	[An, E] [CO4]
11.	Study and interpret Doppler weather radar graphs over any location from the India Meteorological Department (IMD).	[An, E] [CO4]
12.	Analyze any five climate indices and interpret the observations.	[An, E] [CO4]
13.	Plot the tracks of any five cyclone storms over the Indian subcontinent (https://rsmcnewdelhi.imd.gov.in/report.php?internal_menu=MjY) and analyse the observations.	[An, E] [CO4]
14.	Intercompare the rainfall data from numerical weather prediction models over the Indian region with satellite data.	[An, E] [CO4]
15.	Measure atmospheric pressure using different instruments and analyze the data.	[An, E] [CO4]
16.	Measure ground-level ozone concentration and analyze its variation.	[An, E] [CO4]
17.	Measure the surface albedo at different locations and analyze its spatial variation.	[An, E] [CO4]

18.	Analyze aerosol properties using available solar radiometer data.	[An, E] [CO4]
19.	Analyze aerosols using available satellite-based sensor data.	[An, E] [CO4]
20.	Analyse the variation of surface temperature at different locations using infrared thermometer/remote sensing technique.	[An, E] [CO4]
21.	Measure the spectral response of various bodies using a spectrometer.	[An, E] [CO4]
22.	Examine the difference between radiation fluxes computed with and without aerosols using a radiative transfer model (e.g., SBDART).	[An, E] [CO4]
23.	Investigate how surface irradiance depends on the combined effects of cloud optical depth and surface albedo using a radiative transfer model.	[An, E] [CO4]
24.	Compute the spectral surface irradiance in the thermal IR for three cases: clear skies, and high-altitude clouds of optical depth 1 and 5.	[An, E] [CO4]
25.	Estimate the future change in any two atmospheric variables for a given warming scenario using IPCC WG1 Interactive Atlas Simple Climate Futures (https://interactive-atlas.ipcc.ch/).	[An, E] [CO4]